

Wazzup? Building a SOC Lab

Deploying Wazuh from scratch to provide centralized visibility, comprehensive logging, and automated threat response capabilities.

 Ghostface Wazuh Security Overview

Phase 1: Infrastructure & Architecture

4 Core Components

- **Wazuh Indexer:** Stores & indexes alerts for high-speed searching.
- **Wazuh Server:** Decodes data, runs rules, & manages agents.
- **Wazuh Dashboard:** Web interface for visualization.
- **Wazuh Agents:** Installed on endpoints to ship data.

VM Specifications

Spin up VMs (Ubuntu server for manager, Windows/Ubuntu endpoints).

RAM: ~4 GB per VM

CPUs: 2 vCPUs

Disk Space: 50 - 60 GB

Phase 2: Deploying Endpoints

Windows Deployment

Download MSI installer from the dashboard, provide server IP, and execute deployment command via PowerShell as admin.

```
# Run installer with manager address  
msiexec.exe /i wazuh-agent.msi WAZUH_MANAGER="<MANAGER_IP>" /q
```

Linux Deployment & Verification

Select Deb/RPM package, copy script, run via SSH, and confirm agent status is **Active** in the dashboard.

```
wget https://packages.wazuh.com/.../wazuh-agent.deb  
sudo WAZUH_MANAGER="<IP>" dpkg -i wazuh-agent.deb
```

Phase 3: Sysmon Telemetry

Windows & Linux Sysmon

Capture deep process telemetry beyond standard OS logs using robust configuration files (e.g., Olaf Hartong's config).

Log Ingestion Setup

Modify `ossec.conf` on each agent to forward Sysmon event logs to the manager.

```
<localfile>  
  <location>Microsoft-Windows-Sysmon/Operational</location>  
  <log_format>eventchannel</log_format>  
</localfile>
```


Phase 4: Security Modules & Rules

File Integrity Monitoring (FIM)

Watch sensitive directories in real-time by configuring `ossec.conf`:

```
<directories realtime="yes">/etc,/var/www/html</directories>
```

Custom Detection Rules

Add custom XML rules in `local_rules.xml` to catch brute-force attempts or guest account activations.

Phase 5: Dashboards & Active Response

Custom Visualizations

Build custom metrics, line charts, and data tables to track failed logins and security anomalies over time.

Active Response Automation

Trigger automated defenses—such as a firewall-drop script to block attacker IPs during brute-force attacks.

Phase 6: Investigation & Reporting

Validate your SOC lab by simulating attacker actions and reviewing telemetry as a security analyst.

- **Simulate Attacks:** Create/delete users or run discovery commands on endpoints.
- **Analyst Triage:** Investigate captured logs in the Wazuh dashboard.
- **Incident Report:** Compile a report documenting the who, what, when, where, and how of the incident.

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SOC Lab Ready

Your Wazuh infrastructure, endpoints, Sysmon telemetry, and active response rules are fully configured.

Questions & Incident Response Review

Image Sources



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- <https://www.progressive.in/assets/img/photos/SOC-Dashboard.webp?format=webp>

Source: www.progressive.in